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Module 3 CS- Virtualization and Cloud Basics

1. Difference between hardware and software.

Ans: Hardware refers to the physical parts of a computer that can be seen and touched, such as the keyboard, mouse, and monitor.

Software refers to the programs and instructions that tell the hardware what to do, such as Windows and Microsoft Word.

Hardware cannot work without software, and software cannot run without hardware.

In simple terms, hardware is the computer's body, while software is its brain and instructions.

2. Define IP address range and private address range.

Ans: **IP Address Range:**

An IP address range is a group of IP addresses within a specific network that can be assigned to devices for communication. It helps organize and manage network connections.

Private Address Range:

A private address range consists of IP addresses reserved for use within local networks and cannot be accessed directly from the internet. Common private ranges are **10.0.0.0–10.255.255.255**, **172.16.0.0–172.31.255.255**, and **192.168.0.0–192.168.255.255**. These addresses are widely used in homes, schools, and offices.

3. Explain Network protocol and Port number.

Ans: **Network Protocol:**

A network protocol is a set of rules that allows devices to communicate and exchange data over a network. Examples include **HTTP**, **HTTPS**, **FTP**, and **TCP/IP**.

Port Number:

A port number is a logical number used to identify a specific service or application on a device. It helps direct network traffic to the correct program. For example, **HTTP uses port 80** and **HTTPS uses port 443**.

4. Explain Types of Network Devices.

Ans: **Types of Network Devices:**

Network devices are hardware components used to connect and manage communication between computers and networks. Common devices include a **Hub**, which connects multiple devices; a **Switch**, which forwards data to the correct device; a **Router**, which connects different networks; and a **Modem**, which provides internet access. These devices help ensure smooth and efficient data transmission.

5. Which Tools use for Data Backup and Recovery.

Ans: **Tools Used for Data Backup and Recovery:**

Data backup and recovery tools help protect and restore important files in case of data loss.

Common backup tools include **Windows Backup**, **Acronis True Image**, and **Veeam Backup**.

Recovery tools such as **Recuva** and **EaseUS Data Recovery Wizard** can retrieve deleted or lost data. These tools help ensure data safety and business continuity.

6. Explain HTTP and HTTPS Protocols.

Ans: **HTTP and HTTPS Protocols:**

HTTP (HyperText Transfer Protocol) is used to transfer web pages and data between a web browser and a web server. It does not encrypt data, making it less secure.

HTTPS (HyperText Transfer Protocol Secure) is a secure version of HTTP that uses encryption (SSL/TLS) to protect data during transmission. HTTPS helps keep sensitive information, such as passwords and payment details, safe from unauthorized access.

7. What is SSL and TLS Security?

Ans: **SSL and TLS Security:**

SSL (Secure Sockets Layer) and **TLS (Transport Layer Security)** are security protocols used to protect data transmitted over the internet. They encrypt information exchanged between a client and a server, preventing unauthorized access. TLS is the newer and more secure version of SSL. These protocols are commonly used in HTTPS websites to ensure secure communication.

8. Explain the MAC ADDRESS.

Ans: **MAC Address:**

A **MAC (Media Access Control) Address** is a unique identifier assigned to a network device by its manufacturer. It is used to identify devices on a local network. A MAC address is usually a 12-digit hexadecimal number, such as **00:1A:2B:3C:4D:5E**. It helps network devices communicate and transfer data correctly within a network.